

MODEL QUESTION PAPER

TED (15)6091

Reg No.....

REVISION 2015

Signature.....

SIXTH SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY

PLASTIC PRODUCTS

Maximum marks 100

Time 3 hours

(PART -A)

(Maximum marks 10)

I. Answer the following question in one or two sentences. Each question carries 2marks

1. List any four plastic products based on engineering plastics ?
 2. State the importance of flow promoters in thermosetting plastic products with an example ?
 3. List the properties of injection moulded grade HDPE ?
 4. Define SDR ? Write the equations for determination of SDR of PVC pipe ?
 5. Name any four products produced two each by transfer moulding and casting methods ?
- (2x5=10)

(PART -B)

(Maximum marks 30)

II Answer any FIVE questions each question carries 6 marks

1. Explain different factors to be considered while design a plastic product with a typical example
2. State the function with examples of stabilizers, plasticizers, and antidegradants in plastic products.?
3. Mention the characteristics of plastics required for injection molding process with an example?
4. Explain the method of production and raw materials for nylon gears?
5. Explain the manufacture of PVC vinyl flooring with typical formulations.?
6. Describe the production process of disposable plastic glass.?

7. Explain the manufacturing of PMMA sheets?

(5x6=30)

PART -C

(Maximum marks 60)

Answer any one full question from each unit .Each question carry 15 marks

MODULE- I

III a) Describe the various steps involved in the manufacture of plastics products design(8)

b) Explain the rheological behaviour of plastics in product manufacture? (7)

OR

IV a) Mention the environmental impact of plastics product proliferation? (8)

b) Explain the function , properties and applications of fillers and plasticizers used in plastic products with examples (7)

MODULE- II

V a). Explain the manufacture of PVC Chapel with typical formulation (8)

b). Describe the requirements of injection moulds and production methods of bucket (7)

OR

VI a) Explain the process and raw materials for the manufacture of PET bottles(8)

b) Describe the requirements of injection moulds and production methods of chairs?

MODULE- III

VII a) Explain the manufacture of PVC rigid Pipe with typical formulation (8)

b) Explain the manufacture of LDPE blown films (7)

OR

VIII a) Draw and explain line flow chart for rigid PVC pipe extrusion(7)

b) Explain the manufacture of PVC rexin sheets with typical formulations.

MODULE- IV

IX a) Explain the processes Electro plating and Vacuum metalizing of plastic products (8)

b) Describe the manufacture of powder coated plastic products?

OR

X a) Explain the manufacture of inserted plastic products with an example(8)

b) Explain the manufacture of rigid PU foam (7)
